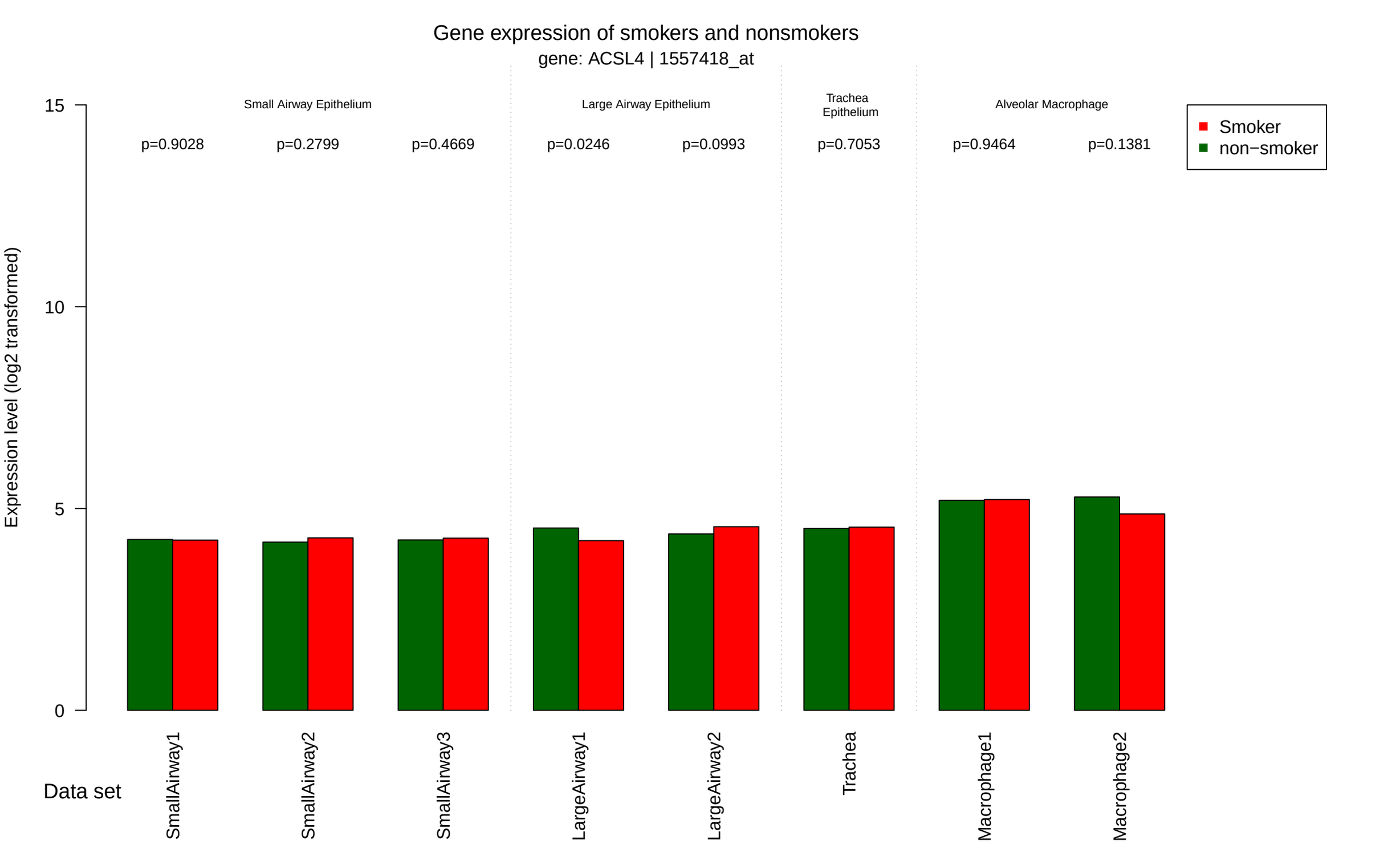
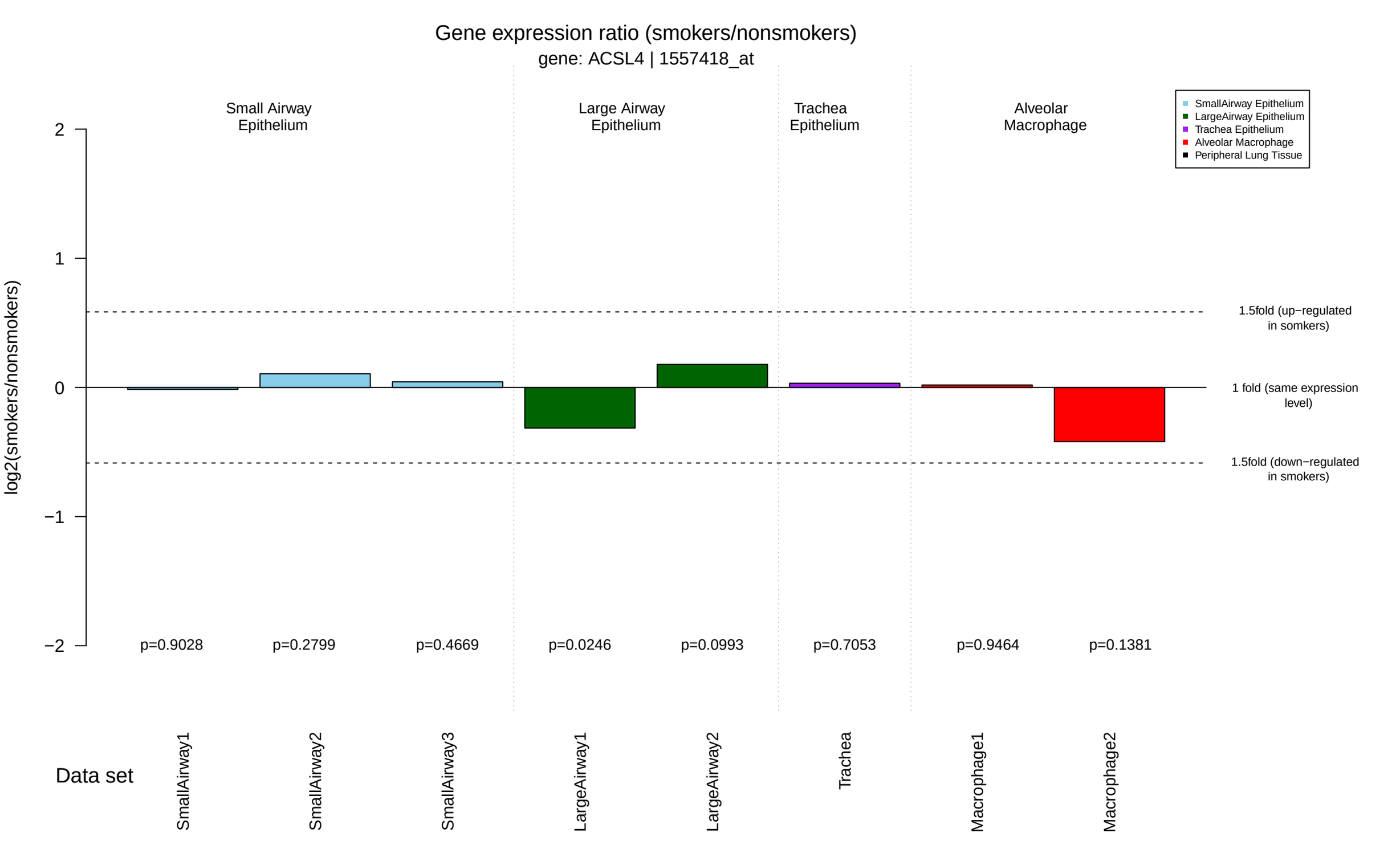
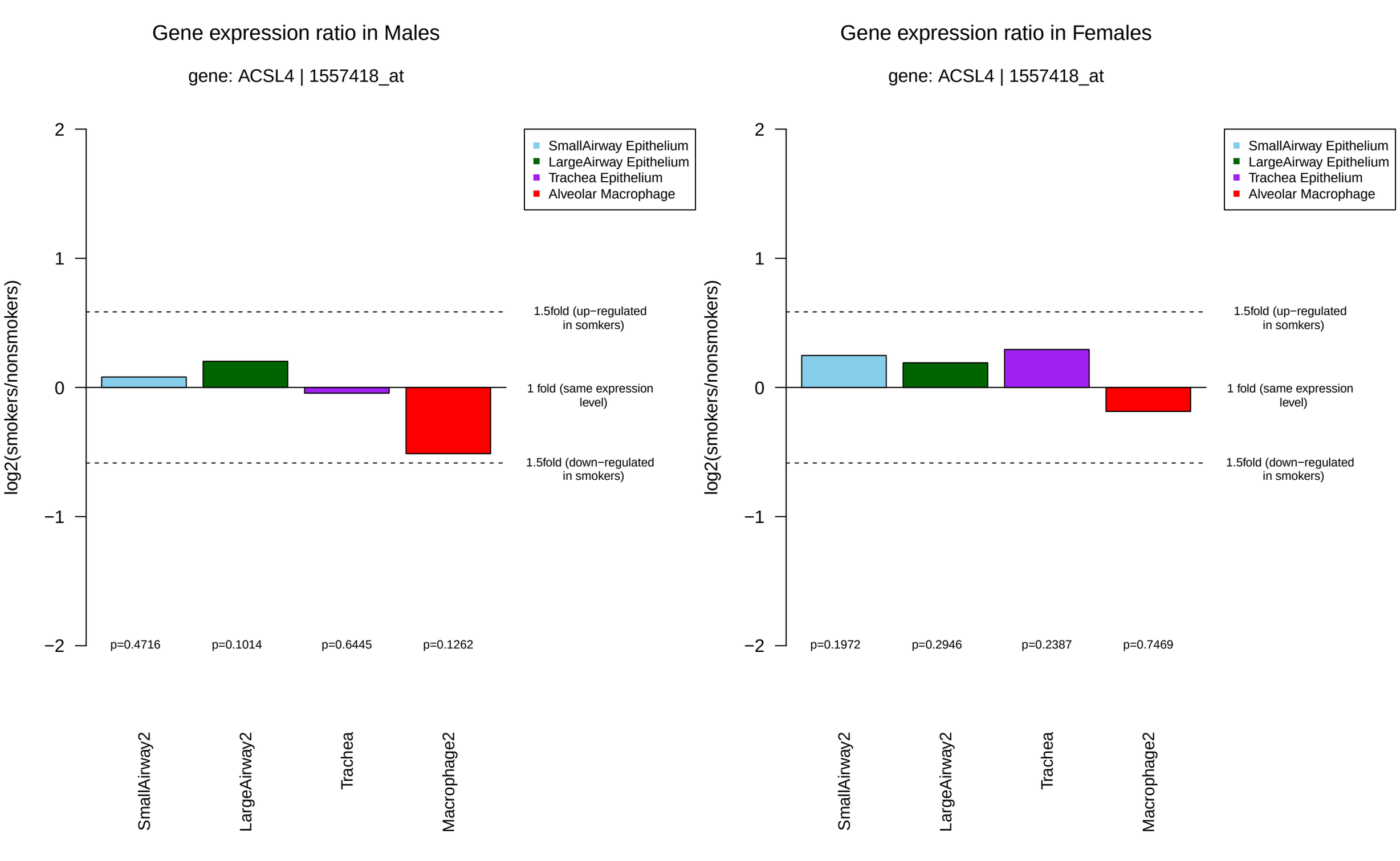
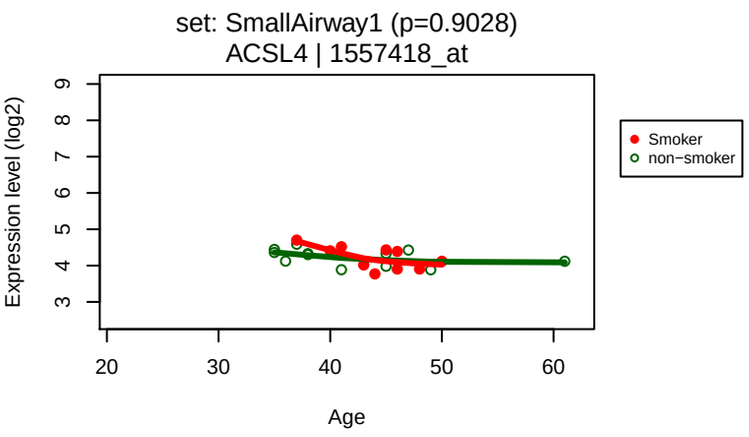
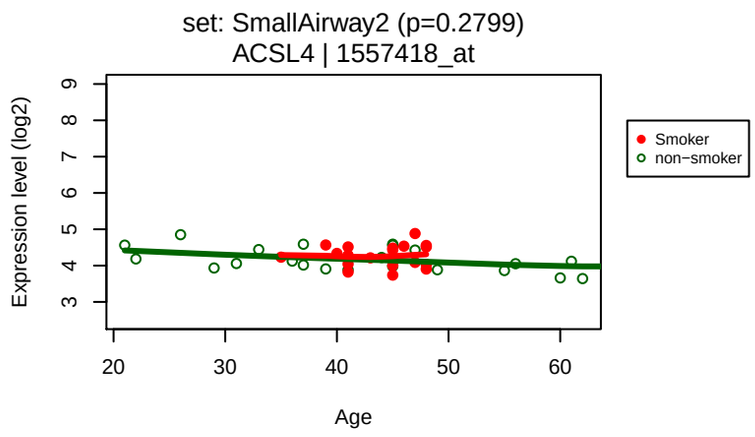
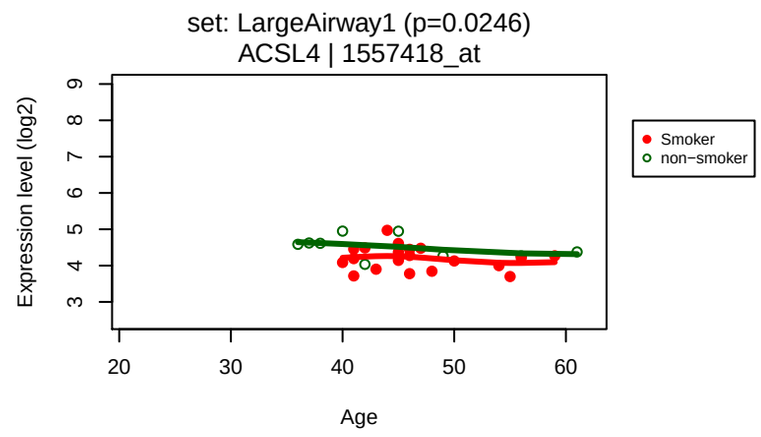

Gene expression of smokers and nonsmokers
gene: ACSL4 | 1557418_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9028
p=0.2799
p=0.4669
p=0.0246
p=0.0993
p=0.7053
p=0.9464
p=0.1381
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

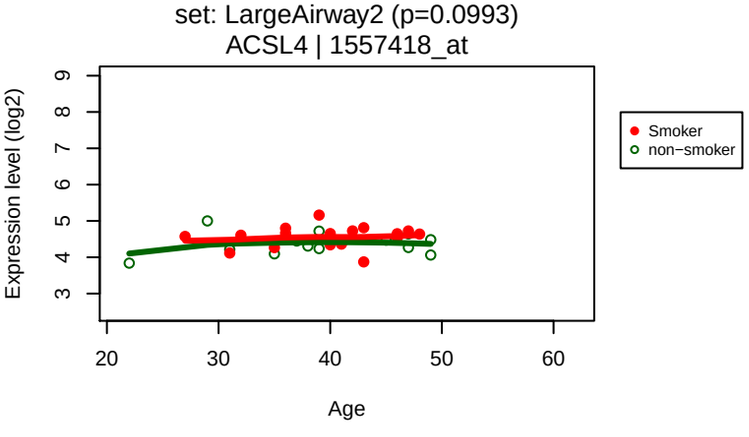

Gene expression ratio (smokers/nonsmokers)
gene: ACSL4 | 1557418_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9028
p=0.2799
p=0.4669
p=0.0246
p=0.0993
p=0.7053
p=0.9464
p=0.1381
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

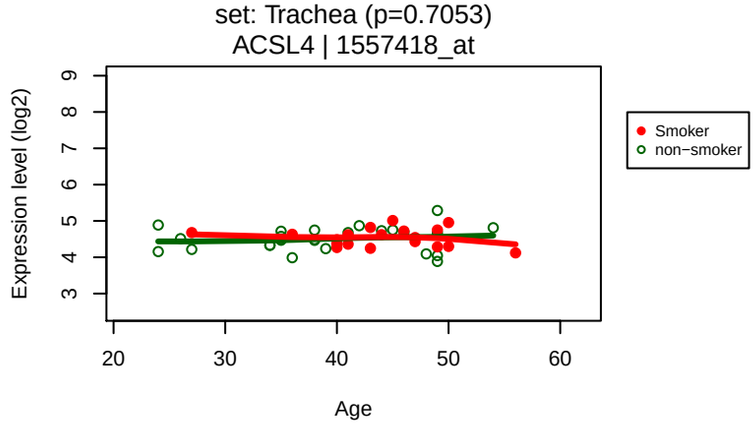

Gene expression ratio in Males
gene: ACSL4 | 1557418_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4716
p=0.1014
p=0.6445
p=0.1262
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSL4 | 1557418_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1972
p=0.2946
p=0.2387
p=0.7469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

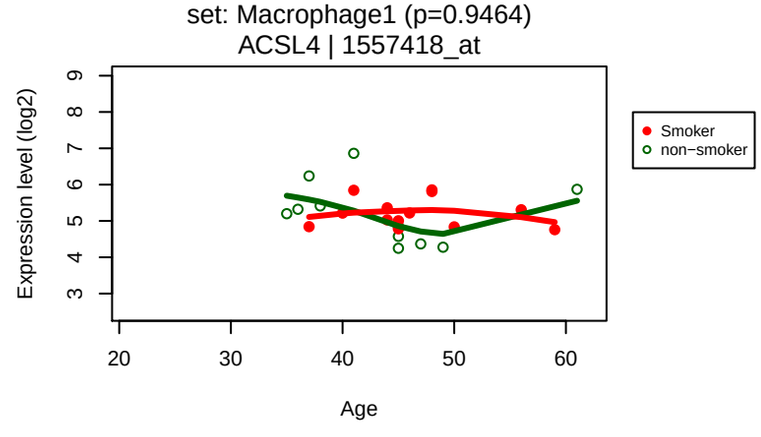

set: SmallAirway1 (p=0.9028)
ACSL4 | 1557418_at
Expression level (log2)
Age
Smoker
non−smoker

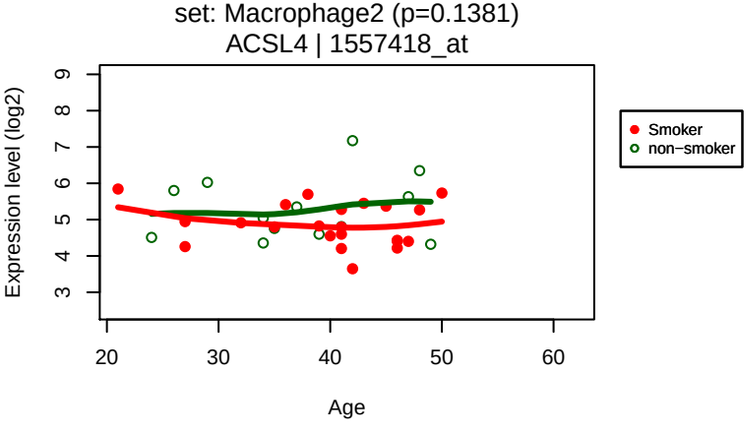

set: SmallAirway2 (p=0.2799)
ACSL4 | 1557418_at
Expression level (log2)
Age
Smoker
non−smoker

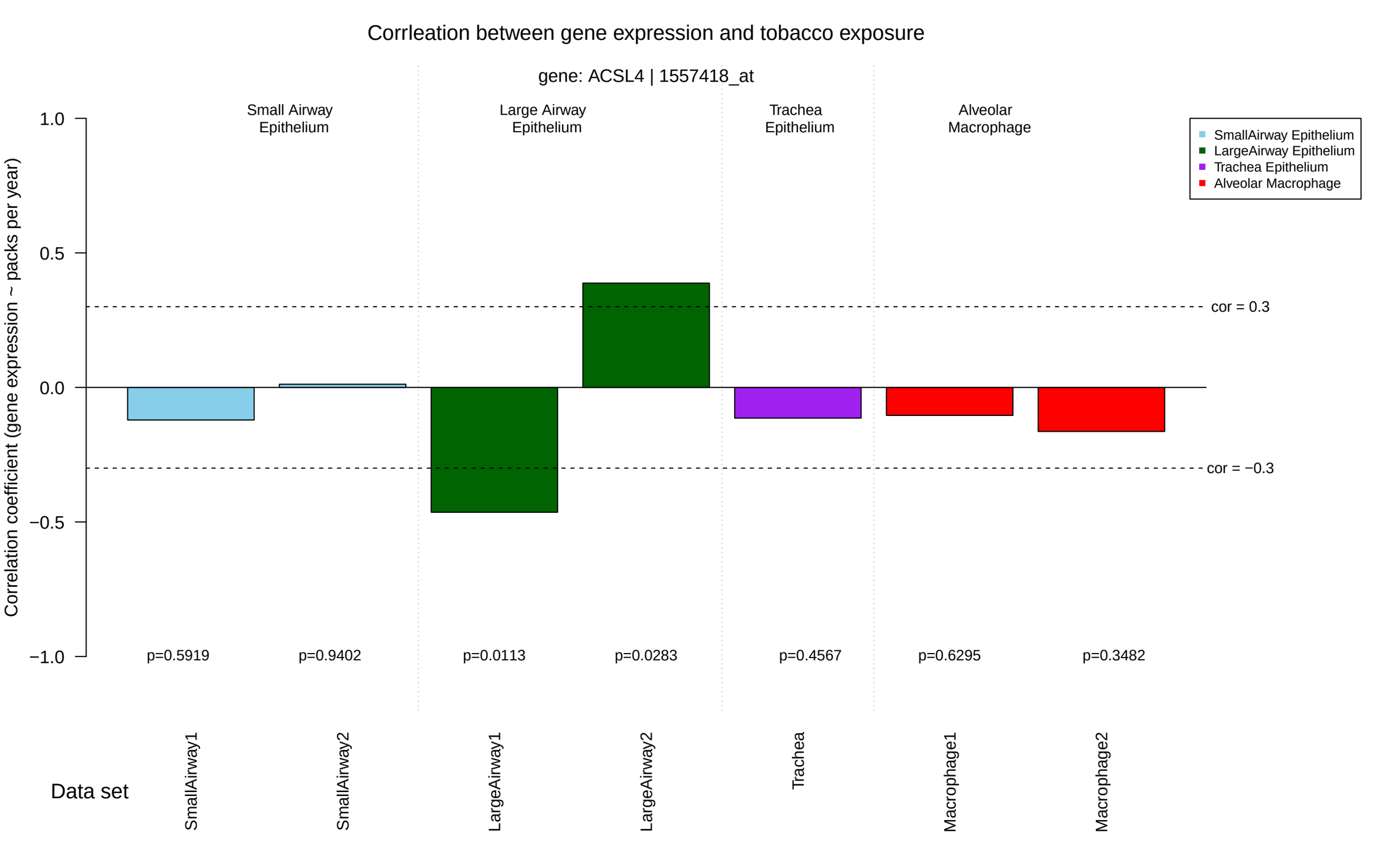

set: LargeAirway1 (p=0.0246)
ACSL4 | 1557418_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0993)
ACSL4 | 1557418_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7053)
ACSL4 | 1557418_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9464)
ACSL4 | 1557418_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1381)
ACSL4 | 1557418_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSL4 | 1557418_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5919
p=0.9402
p=0.0113
p=0.0283
p=0.4567
p=0.6295
p=0.3482
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2